FPGA Programming – Comprehensive Syllabus

Course Overview

This course introduces Field Programmable Gate Arrays (FPGAs) and digital hardware design using Hardware Description Languages (HDLs). Students will learn digital design fundamentals, FPGA architecture, Verilog/VHDL programming, simulation, synthesis, timing analysis, hardware interfacing, and real-time FPGA application development.

Duration: 12–16 Weeks (60–80 Hours)

Prerequisites:

- Basic Digital Electronics
- Logic Gates and Boolean Algebra
- Basic Computer Architecture (recommended)
- No prior FPGA experience required

Module 1: Introduction to FPGA & Digital Design (Week 1)

Topics

- Introduction to FPGA
- FPGA vs Microcontroller vs ASIC
- FPGA Architecture
- Configurable Logic Blocks (CLBs)
- Look-Up Tables (LUTs)
- Flip-Flops
- Routing Resources
- FPGA Applications
- Development Tools Overview

Practical Exercises

- FPGA Board Setup
- Development Environment Installation
- First FPGA Project

Module 2: Digital Logic Design Fundamentals (Week 2)

Topics

- Number Systems
- Boolean Algebra
- Logic Gates
- Combinational Logic
- Sequential Logic
- Karnaugh Maps (K-Maps)
- Multiplexers
- Demultiplexers
- Encoders & Decoders

Practical Exercises

- Logic Gate Implementation
- Multiplexer Design
- Seven-Segment Display Decoder

Module 3: Verilog HDL Fundamentals (Week 3)

Topics

- Introduction to Verilog
- Module Structure
- Data Types
- Operators
- Continuous Assignment
- Procedural Blocks
- Conditional Statements
- Loops
- Functions
- Tasks

Practical Exercises

- Basic Verilog Programs
- Arithmetic Logic Unit (ALU)
- Combinational Circuits

Module 4: Sequential Circuit Design (Week 4)

Topics

- Flip-Flops
- Registers
- Counters
- Shift Registers
- Finite State Machines (FSM)
- Mealy & Moore Machines

Practical Exercises

- Binary Counter
- Ring Counter
- Traffic Light Controller

Module 5: Simulation & Verification (Week 5)

Topics

- Testbenches
- Simulation Concepts
- Waveform Analysis
- Functional Verification
- Debugging HDL Code

Practical Exercises

- ALU Simulation
- Counter Verification
- FSM Testing

Module 6: FPGA Design Flow (Week 6)

Topics

- Design Entry
- Synthesis
- Place and Route
- Bitstream Generation
- FPGA Configuration
- Timing Analysis
- Resource Utilization

Practical Exercises

- Complete FPGA Design Flow
- Downloading Bitstream
- Timing Report Analysis

Module 7: FPGA I/O Interfacing (Week 7)

Topics

- GPIO
- LEDs
- Push Buttons
- Switches
- Seven-Segment Displays
- LCD Interface
- VGA Basics

Practical Exercises

- LED Blinking
- Button Interface
- Seven-Segment Counter

Module 8: Memory Design (Week 8)

Topics

- ROM
- RAM
- FIFO
- Registers
- Block RAM (BRAM)
- Memory Controllers

Practical Exercises

- RAM Design
- FIFO Implementation
- Memory Read/Write Operations

Module 9: Communication Interfaces (Week 9)

Topics

- UART
- SPI
- I²C
- Ethernet Basics
- USB Basics
- High-Speed Serial Interfaces

Practical Exercises

- UART Transmitter & Receiver
- SPI Communication
- I²C Master Interface

Module 10: Advanced FPGA Design (Week 10)

Topics

- Clock Management
- PLL
- Clock Domain Crossing
- Pipelining
- Parallel Processing
- DSP Blocks
- Hardware Optimization

Practical Exercises

- Clock Divider
- Pipelined Adder
- DSP-Based Arithmetic

Module 11: FPGA-Based Embedded Systems (Week 11)

Topics

- Soft Processors
- Embedded Processors
- System-on-Chip (SoC)
- FPGA with ARM Processors
- Hardware/Software Co-Design

Practical Exercises

- Soft-Core Processor Demo
- Peripheral Interfacing
- SoC-Based Application

Module 12: Mini Project (Week 12)

Choose one:

- Digital Clock
- Elevator Controller
- Traffic Light Controller
- UART Communication System
- Digital Calculator

- Electronic Voting Machine
- Frequency Counter

Module 13: Capstone Project (Week 13–16)

Complete FPGA System Development

- Requirement Analysis
- HDL Design
- Simulation
- Synthesis
- Timing Optimization
- Hardware Implementation
- Testing & Debugging
- Documentation
- Final Presentation

Tools & Software

- Xilinx Vivado
- Intel Quartus Prime
- ModelSim
- QuestaSim
- GTKWave
- Verilog HDL
- VHDL (optional)

Hardware Platforms

- Xilinx Artix-7 FPGA
- Xilinx Spartan-6 FPGA
- Intel Cyclone IV/V FPGA
- Basys 3 Development Board
- Nexys A7 Development Board
- DE10-Lite FPGA Board

Assessment Scheme

Component	Weightage
Assignments	20%
Lab Exercises	20%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of the course, learners will be able to:

- Understand FPGA architecture and programmable logic concepts.
- Design combinational and sequential digital circuits using Verilog HDL.
- Simulate, verify, and debug FPGA designs.
- Implement memory modules, counters, FSMs, and communication interfaces.
- Perform synthesis, place-and-route, and timing analysis.
- Interface FPGA boards with external devices and peripherals.
- Develop optimized FPGA-based embedded systems.
- Build complete FPGA applications from design to hardware implementation.
- Apply FPGA design techniques in industries such as telecommunications, automotive, aerospace, robotics, consumer electronics, and industrial automation.